Forecasting neighborhood activity and identifying hotspot candidates

01 Observed history

Mobility history

Past neighborhood arrival activity

Business dynamics

New licenses and closures, lagged

Context indicators

Observed subway or living-population measures

02 Forecasting model

ARIMAX / SARIMAX

monthly neighborhood series

Reference models: naïve · seasonal naïve · ARIMA / SARIMA

03 Future activity

Predicted mobility indicator

$$\hat{y}_{d,\,t+1:t+3}$$

Next three monthly observations

04 Rule-based hotspot assessment

Activity level

How active is the area?

Growth

Change from the comparable period

Persistence

Is the change sustained?

Output **Ranked hotspot candidates by administrative dong** →

Only information available at the forecast date enters the model; assessment criteria are fixed using development data.